Technical Section — Internal Draft

Equivariant Neural Networks on MNIST

4-Model Comparison & the 6/9 Problem

PART 1 · 4-MODEL EXPERIMENT

Setup and Main Result

1. Setup

Train 4 models, and evaluate on unrotated MNIST / rotated MNIST.

- CNN → trained on unrotated MNIST / trained on rotated MNIST
- C_4 ENN → trained on unrotated MNIST / trained on rotated MNIST

Each of the 4 trained models is then evaluated on **both** unrotated MNIST and rotated MNIST.

Definitions

C_4 ENN C_4: the C_4 group. ENN: Equivariance Neural Network.

MNIST A dataset of handwritten-digit pictures, each equipped with the label labeling the digit shown in the picture. Both train and test data are included.

rotated Before train/test, the picture was rotated in advance.

Evaluation procedure

Evaluate the trained model: test the accuracy of correctly distinguishing the digit given a test picture.

> **Footnote — reason for C_4** For the grid $\mathbb{Z}^2$, only subgroups of D_4 are exact symmetries. (ref: chap. 2.8)

▸ **NOTE TO REPORTER**

These parts are necessary background — in the actual report, some of these parts can be omitted accordingly.

2. Result

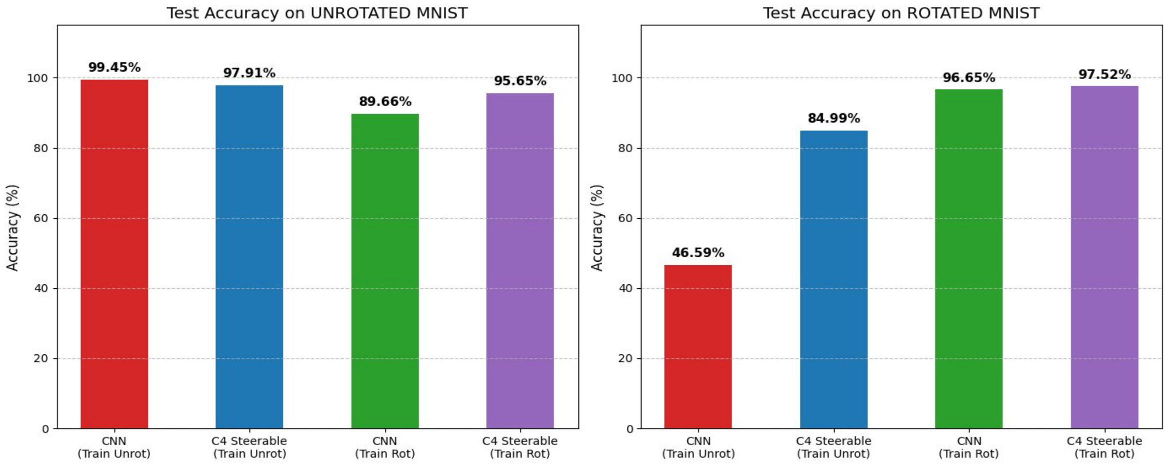

Figure 1. Test accuracy on unrotated MNIST (left) and rotated MNIST (right) for all 4 trained models.

△ Main result

① Compare CNN (train rot) and C_4 ENN (train rot) on rotated MNIST:

CNN → **96.65%** vs C_4 ENN → **97.52%**

We find C_4 steerable has the better accuracy, which means the inherited rotational symmetry built into the network does make models perform better at distinguishing rotated data.

> ▸ **NOTE TO REPORTER**
> *> Other comparisons — these will be too long for a 5-min report. Only listed for completeness of discussion; in practice one can skip or quickly go through this part.*

Other comparisons

② Unrot CNN, unrot C_4 ENN → tested on rotated MNIST:

CNN → **46.59%** vs C_4 ENN → **84.99%**

CNN performs awful, which means it learns almost nothing about the rotated feature of the digit. C_4 ENN still performs pretty good, which indicates it doesn't learn rotation from the data, but inherits the rotational symmetry by construction. (For more info, refer to chap. 2.)

③ Unrotated CNN, unrot C_4 ENN → tested unrotated:

CNN → **99.45%** vs C_4 ENN → **97.91%**

The compulsory group restriction applied on C_4 ENN leaves it with fewer learnable parameters, making it in contrast less expressive on the standard task.

④ Rot CNN, rot C_4 ENN → tested unrotated:

CNN → **89.66%** vs C_4 ENN → **95.65%**

Benefiting from a built-in symmetry structure, ENN can put more resource into learning the digit itself rather than the rotated feature. CNN has to separate some resource to learn the rotated feature by rote (*Capacity Dilution*).

PART 2 · A PITFALL OF ROTATIONAL ENN

The 6/9 Problem and Group Restriction

1. Introduction

The ENN suffers from the 6/9 problem.

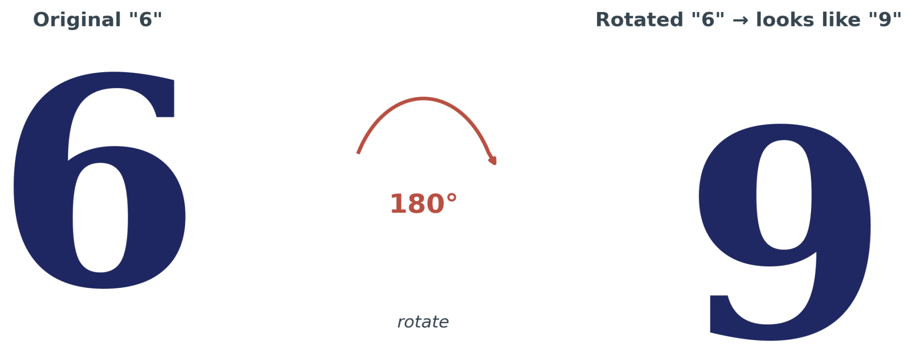

Figure 2. A handwritten “6” rotated by 180° becomes visually indistinguishable from a “9”.

- A rotationally-symmetric ENN can only learn *in the sense of rotation*.
- In the sense of rotation, 6 and 9 look alike.
- Therefore, it is easy for an ENN to mistake 6 for 9.

> **Footnote** The 6/9 problem generalizes to other digits (for instance 5 and 7), to other tasks beyond telling digits, and to broader symmetry groups beyond rotation.

2. Solution: Group Restriction

In this case, we first follow the same C_4 structure as the traditional C_4 ENN in the first 5 layers (i.e. layers are invariant under the C_4 group operation — 90°, 180°, 270°, 360° rotation).

Then we restrict the group in the sixth layer: $C_4 \to \{e\}$. The sixth layer is only required to be invariant under the trivial group $\{e\}$ (i.e. no symmetry restriction).

Architecture comparison

① Traditional C_4 ENN

input → **L_1** → **L_2** → **L_3** → **L_4** → **L_5** → **L_6 (out)**

input ⇔ output : C_4 operation invariant

② Refined $C_4|_{\{e\}}$ ENN

input → **L_1** → **L_2** → **L_3** → **L_4** → **L_5** → **L_6 (out)**

input ⇔ L_5 : C_4 operation invariant | L_5 ⇔ L_6 : $\{e\}$ operation invariant (group restriction $C_4 \to \{e\}$)

> **Footnote ①** In practice we also need to claim the representation of the group, and there is a fully-connected classifier at the end. Here for simplicity we omit those parts.

> **Footnote ②** For details on group restriction, see ref chap. 2.7.

3. Intuitive Perspective

The deeper a layer is, the broader its receptive field is. Therefore, requiring different symmetries between layers of different depth is actually equivalent to requiring different symmetries at different scales. In this $C_4|_{\{e\}}$ design we sacrifice the large-scale symmetry while maintaining the small-scale C_4 symmetry. (For further info, refer to chap. 2.7.)

▸ **NOTE TO REPORTER**
For timing: the reporter can selectively skip or quickly go through the technical and the understanding parts. Focus on the intro and the result.

KEY POINT
Group restriction is a technique to mitigate the 6/9 problem.

4. Result

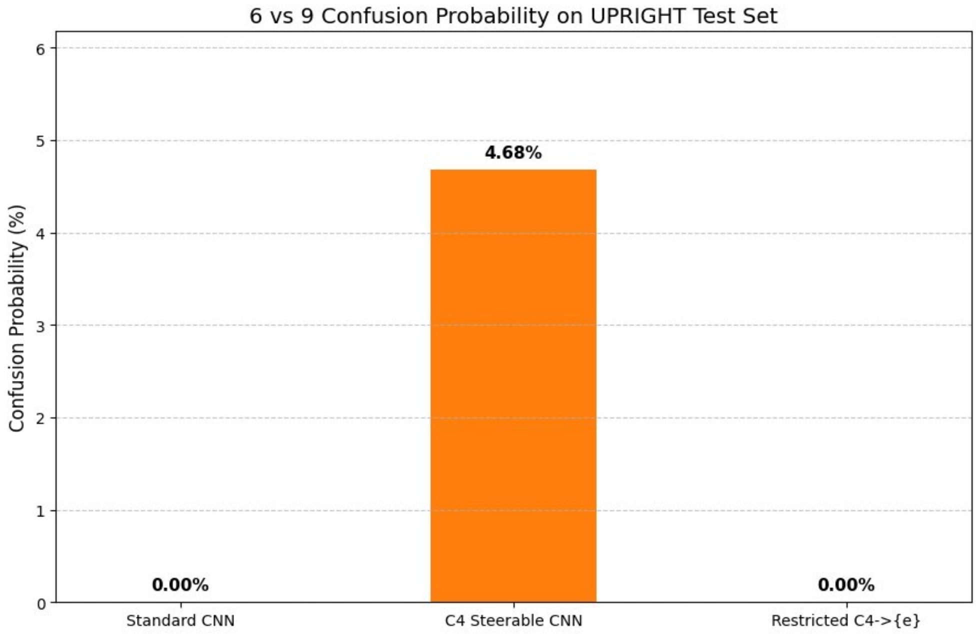

Figure 3. 6-vs-9 confusion probability on the upright test set.

Clearly, the traditional C_4 ENN suffers from the 6/9 problem, while $C_4|_{\{e\}}$ and CNN don't have the problem.

Footnote However, the 6/9 problem in traditional C_4 ENN is not so severe in practice, since handwritten 6 and 9 still have differences even in the sense of rotation.

REFERENCES

Source

All chapter references in this draft point to:

General E(2)-Equivariant Steerable CNNs

Maurice Weiler, Gabriele Cesa · NeurIPS 2019

https://arxiv.org/abs/1911.08251

Chapters cited in this draft

- **Chapter 2.8** — rationale for choosing the C_4 group on the $\mathbb{Z}^2$ grid
- **Chapter 2** — inherent rotational features of C_4 ENN constructions
- **Chapter 2.7** — full details of group restriction